

Module 13: Case Study of Oracle Concurrency Control

AKTU BCS501 | Unit 5: Concurrency Control Techniques

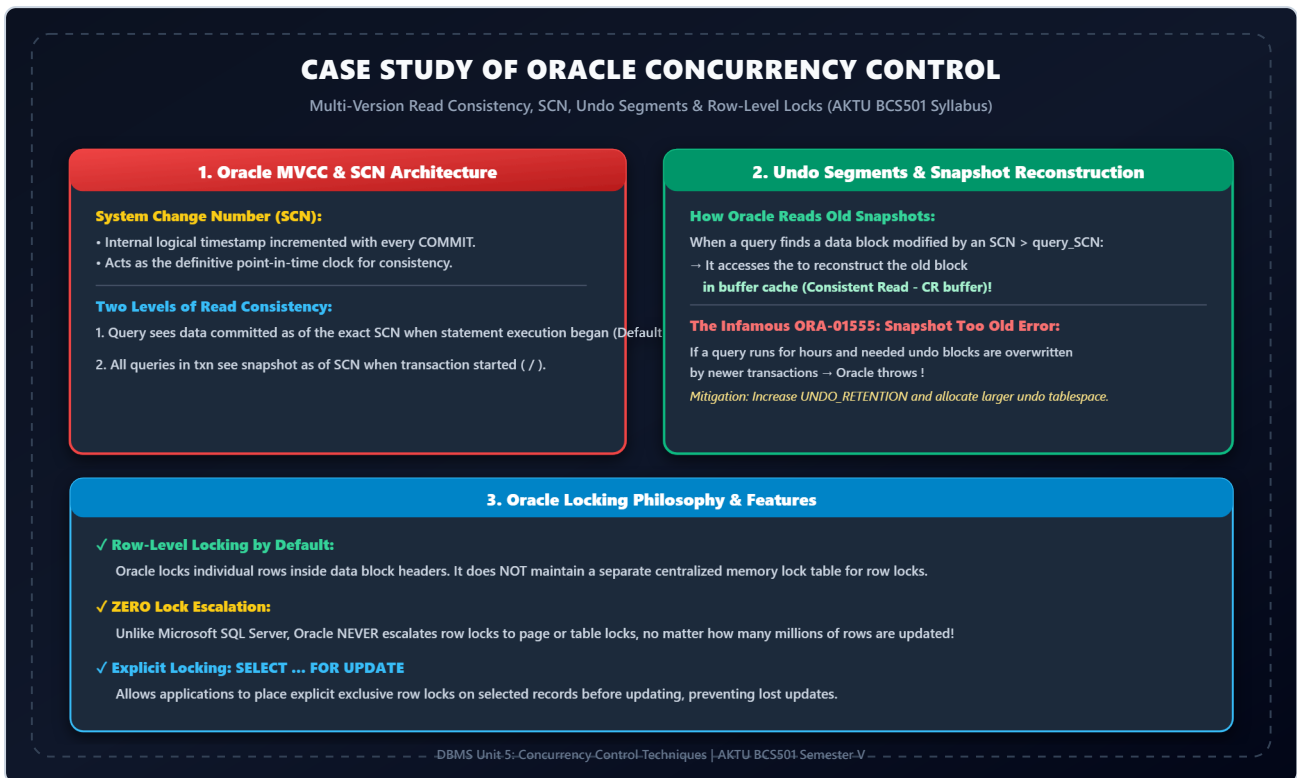


Figure 13.1: Oracle MVCC, System Change Number (SCN) & Undo Segment Reconstruction

1. Oracle Multi-Version Read Consistency

Oracle never allows readers to block writers or writers to block readers. Readers access historical snapshots reconstructed from Undo Segments based on the query's start SCN.

2. Key Oracle Features

- **SCN (System Change Number):** Logical clock incremented at every commit.
- **ORA-01555 (Snapshot Too Old):** Occurs when required undo blocks are overwritten.
- **No Lock Escalation:** Row locks are stored in block headers and never escalate to table locks.